

Differential Geometry Assignment 2

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Definition 1. The n -manifold $S^n(r)$ is defined as $\{v \in \mathbb{R}^{n+1} : \|v\| = r\}$

1 Vector fields and reparametrization

Claim 1. Suppose γ is a smooth curve on $[a, b]$, V is a parallel vector field along γ and $\phi : [c, d] \rightarrow [a, b]$ is a diffeomorphism. Then $V \circ \phi$ is a parallel vector field along $\gamma \circ \phi$

Proof. We have $(V \circ \phi)'(t) = V'(\phi(t))\phi'(t)$. We also have $V'(\phi(t)) \in T_{\gamma(\phi(t))}M$ since V is parallel to γ . $\square$

2 Tangential and normal frames

Definition 2. We define $\mathcal{Y}(M) = \{f \in C^\infty(M, \mathbb{R}^k) : f(p) \in N_p M\}$

Definition 3. A tangential frame on a neighborhood V of p consists of $X_i \in \mathcal{X}(V)$ such that $X_i(q)$ form an orthonormal basis of $T_p M$. A normal frame is defined similarly.

Claim 2 (Gram-Schmidt). Suppose we have $\tilde{X}_i \in C^\infty(M, \mathbb{R}^k)$ such that $\tilde{X}_i(q)$ are linearly independent. Let $X'_1 = \tilde{X}_1$ and define X'_{k+1} recursively by

$$X'_{k+1} = \tilde{X}_{k+1} - \sum_{i=1}^k \frac{\langle X'_i, \tilde{X}_{k+1} \rangle}{\langle X'_i, X'_i \rangle} X'_i$$

Now define $X_i = \frac{X'_i}{\|X'_i\|}$. Then $X_i(q)$ are orthonormal and have the same span as $\tilde{X}_i(q)$

Claim 3. There is a tangential frame on some neighborhood V

Proof. Apply the Gram-Schmidt process to $\phi_i \in \mathcal{X}(V)$ obtained from a local parametrization $\phi \in C^\infty(\Omega, V)$ $\square$

Claim 4. There is a normal frame on some neighborhood V

Proof. Define $Y_i \in \mathcal{Y}(M)$ by $Y_i(q) = \hat{\pi}_q(e_i)$ and

$$V = \{q : Y_i(q) \text{ are linearly independent}\}$$

Use the determinant function to see that V is open. Finally, apply Gram-Schmidt. $\square$

Claim 5. Suppose $N_i \in \mathcal{Y}(V)$ are a normal frame on V . Then

$$S_p(v, w) = - \sum_{i=1}^{k-n} \langle (D_v N_i)(p), w \rangle N_i(p)$$

Proof. Get $\gamma \in C^\infty((-\epsilon, \epsilon), M)$ with $\gamma(0) = p$ and $\gamma'(0) = v$. We have

$$\langle N_i(\gamma(t)), \pi_{\gamma(t)} w \rangle = 0$$

Differentiating the map $t \rightarrow \pi_{\gamma(t)} w$ at 0 we get $[(D\pi)_p v]w = S_p(v, w)$ and differentiating the map $t \rightarrow N_i(\gamma(t))$ at 0 we get $(D_v N_i)(p)$. Now using the rule for differentiating inner products we get the desired result. $\square$

Claim 6. For the manifold $S^n(r)$ we have $S_p(v, w) = -\frac{\langle v, w \rangle}{r^2} p$

Proof. Define $\gamma(t) = r \frac{p+tv}{\|p+tv\|}$ and $N(p) = \frac{p}{r}$. Then $(N \circ \gamma)'(0) = \frac{v}{r}$. Now use the previous claim. $\square$

3 Local Riemannian isometry

Suppose $F \in C^\infty(M, N)$ is a local Riemannian isometry.

- For $\gamma \in C^\infty(I, M)$ define $\tilde{\gamma} \in C^\infty(I, N)$ by $F \circ \gamma$
- For V in $\mathcal{X}(\gamma)$ define $\tilde{V} \in \mathcal{X}(\tilde{\gamma})$ by $t \rightarrow DF_{\gamma(t)}(V(t))$

Claim 7. We have $DF_{\gamma(t)}[cov V(t)] = cov \tilde{V}(t)$ for all $t \in I$

Proof. Choose a local parametrization $\phi \in C^\infty(\Omega, V)$ for a neighborhood V of $p = \gamma(t_0)$ such that $F|_V \in C^\infty(V, F(V))$ is a Riemannian isometry and $F(V)$ is open. Using formula (4.15) from Chapter 4 of [LeeRM] the left hand side becomes

$$\dot{V}^j(F_* \phi_j)(\tilde{\gamma}(t)) + V^j[DF_{\gamma(t)}(\nabla_{\gamma'(t)} \phi_j)] = \dot{V}^j(F_* \phi_j)(\tilde{\gamma}(t)) + V^j[\nabla_{\tilde{\gamma}'(t)}(F_* \phi_j)]$$

We can expand the right hand side as

$$cov[t \rightarrow V^j(F_* \phi_j)(\tilde{\gamma}(t))](t)$$

Applying (4.15) again we see that this is the LHS $\square$

Claim 8. If γ is a geodesic then so is $\tilde{\gamma}$

Proof. Substitute $V = \gamma'$ in the previous claim. $\square$

Claim 9. *We have $DF_{\gamma(t_2)}P_{t_1t_2}^\gamma = P_{t_1t_2}^{\tilde{\gamma}}DF_{\gamma(t_1)}$*

Proof. Let $v \in T_{\gamma(t_1)}M$. Let $V \in \mathcal{X}(\gamma)$ be parallel to γ with $V(t_1) = v$. Applying the left hand side to v we get $\tilde{V}(t_2)$. We also have $\tilde{V}(t_1) = DF_{\gamma(t_1)}v$. Hence applying v to the RHS we also get $\tilde{V}(t_2)$ $\square$